LAB1: Odometry

On a mobile robot, implement a communication protocol of your own design, based on a serial interface, which:

1. allows the robot to be commanded to drive straight forward or backward by a given distance;
2. allows the robot to turn right or left by a given angle (with a resolution of at least 45 degrees).

The robot's position must be determined from the encoders on the robot's drive motors, and control must run in a feedback loop that improves the accuracy of the executed movements.

The final test is for the robot to complete the following task:

1. The operator drives the robot manually from a given point A to point B.
2. The robot is then carried back to point A. A sequence of commands is sent to it over the serial interface, and with these commands the robot drives from point A to point B autonomously.

Good luck
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